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Free-space optical (FSO)-based satellite communications, due to extremely high data rates and global coverage capability, have recently drawn substantial research attention. The adverse issues on FSO-based satellite links, including atmospheric turbulence and pointing error, pose various challenges in designing and deploying such systems. Nevertheless, recent efforts in error control design focusing on adaptation-based mitigation techniques face practical restrictions due to the outdated feedback channel state information (CSI) caused by long-distance/high-latency satellite links. This article addresses the design of an adaptive rate/power control scheme using machine learning (ML)-aided channel prediction for FSO-based satellite systems. Notably, we employ the echo state network (ESN) model, an efficient form of recurrent neural network, for channel prediction. The design proposal facilitates the concurrent control of data rate and satellite's transmitted power for each equal-duration channel state, leveraging accurately predicted CSI. The average required transmitted power and energy efficiency performance metrics are analytically derived. Numerical results demonstrate the severe impact of outdated CSI on the performance of FSO-based satellite systems and highlight the necessity of our design proposal. Moreover, we confirm the effectiveness of the ESN model for FSO channel prediction by comparing its performance with other ML approaches in terms of predicted accuracy and complexity.

I. INTRODUCTION

Recent years have witnessed the growing trend to provide Internet services from low earth orbit (LEO) satellites [1]. The revolutionary impact of thousands of LEO satellites in projects, such as SpaceX's Starlink, Amazon's Kuiper, Telesat, and OneWeb, has triggered a surge in global mega satellite-constellation development [2]. On the other hand, free-space optical (FSO) communication has gained renown for providing high-speed data services over long distances without depleting radio frequency (RF) resources [3]. Leveraging the recent progress in satellite-access Internet and full-fledged hardware facilitating FSO technology, FSO-based LEO satellite communication has emerged as a potential solution for diverse applications, e.g., vertical fronthaul/backhaul networks, last-mile access

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of the Internet of Vehicles (IoV), and quantum key distributions [4]. However, the impact of atmospheric turbulence and pointing errors on FSO links significantly degrades system performance, necessitating extensive research efforts [5].

Related works: Extensive research endeavors have been steered to discover efficient mitigation techniques for FSO-based LEO satellite systems [3]. These studies primarily focused on: 1) physical-layer (PHY) solutions, e.g., hybrid FSO/RF scheme [5], adaptive transmission [6], and intelligent-reflecting-surface-aided relays [7], and 2) link-layer solutions, e.g., hybrid automatic repeat request [8]. A promising and viable approach to further improve the system performance is the joint design between the adaptive transmission scheme with other mitigation techniques, which has recently gained considerable attention [4], [6], [7], [8]. Using the adaptive transmission scheme, the transmitter adaptively selects the appropriate PHY parameters, such as data rate, coding rate, and transmitted power, to maintain the quality of service (QoS) requirements over time-varying turbulence channels [9], [10], [11], [12], [13], [14]. Notably, Perlot and de Cola [9] investigated the throughput performance of optical LEO satellite-to-ground systems using adaptive rate transmissions. Spellmeyer et al. [10] analyzed the performance of multirate differential phase-shift keying schemes for optical satellite systems. The performance of optical LEO satellite systems using adaptive pulse position modulation was investigated in [11]. Geisler et al. [12] conducted an experiment to demonstrate the feasibility of employing adaptive modulation (AM) and coding rate schemes for coherent optical LEO satellite systems. German researchers presented the implementation of adaptive rate schemes, where LEO satellites adjust the data rate according to elevation and channel conditions [13]. Most recently, the novel architecture to implement variable data rate schemes in the hardware efficiently was introduced in [14]. In this work, the authors aimed to achieve the highest symbol rate allowed by the link budget in each sector of the satellite pass by repeating/spreading the data symbols while keeping the targeted chip rate.

Motivation: Existing works on FSO-based LEO satellite systems using adaptive transmission schemes also leave some open challenges and issues for future research. *First*, the aforementioned studies primarily focused on the design of data rate adaptation based on a fixed transmitted power selected for the worst case channel conditions. Using a constant transmitted power over time-varying turbulence channels leads to low achievable energy efficiency, which is especially critical for satellite communications. Rate adaptation in conjunction with transmitted power control can take advantage of favorable channel conditions, resulting in considerable enhancement in energy efficiency without sacrificing the desired QoS, e.g., targeted bit error rate (BER) [15]. It is worth noting that the design of joint rate and power adaptation has been extensively investigated for FSO-based terrestrial systems [16], [17], [18]. In these studies, the transmitted power was adjusted in the order of bit duration to strictly maintain the targeted BER. This,

TABLE I
Literature Comparison on Optical LEO Satellite Systems Using Adaptive Transmission Schemes

Works	Adaptive		Vehicular Networks	Outdated CSI	Channel Prediction
	Rate	Power			
[4], [6], [7]	✓	×	✓	×	×
[8], [9], [10], [11], [12], [13], [14]	✓	×	×	×	×
Our study	✓	✓	✓	✓	✓

nonetheless, poses challenges in practical systems due to feedback delays and hardware limitations, particularly in long-distance satellite communications. Therefore, a proper adaptive rate/power control design for optical LEO satellite systems is required.

Regarding the second concern, the existing design of adaptive transmission schemes for FSO-based LEO satellite systems was based on the assumption of perfect channel state information (CSI). In practice, the CSI tends to be outdated, especially in optical satellite systems. Unlike FSO-based terrestrial systems,¹ this issue becomes critical due to inherent high feedback latency in long-distance LEO satellite communications, i.e., a few milliseconds. This delay is much longer than the LEO satellite channel coherence time, which is typically shorter than 1 ms, as reported in [22]. Such outdated CSI, indeed, substantially deteriorates the performance of optical satellite systems using adaptive transmission schemes [23]. Therefore, it becomes crucial and imperative to devise an efficient channel prediction for accurately acquiring the up-to-date CSI over time-varying FSO channels. Our initial report in [24] confirmed the effectiveness of machine learning (ML)-based echo state network (ESN) for FSO channel prediction. The ESN model, an efficient recurrent neural network (RNN), constructs random recurrent connections within its hidden layer and utilizes a straightforward linear regression algorithm for training the output layer [25]. This model overcomes the limitations of traditional RNNs, i.e., elevated computational complexity and slow convergence, making it a potential candidate for channel prediction [26]. To our knowledge, the design of adaptive rate/power control with ML-based channel prediction is not available in the literature on FSO-based LEO satellite systems.

Contributions: The primary objective of this article is to present the efficient design of adaptive rate/power control with ML-based channel prediction for FSO-based LEO satellite-assisted vehicular networks. A comparison between our study and the literature is depicted in Table I. In summary, the key contributions of this article are outlined as follows.

¹In FSO-based terrestrial systems, the impact of outdated CSI is not significant. This is because the link distance is a few kilometers, in which the delay of CSI estimation based on the feedback channel is short enough [19], [20]. On the other hand, thanks to the terrestrial channel reciprocal characteristics, the CSI can be estimated for transmitting channels using the received CSI [21].

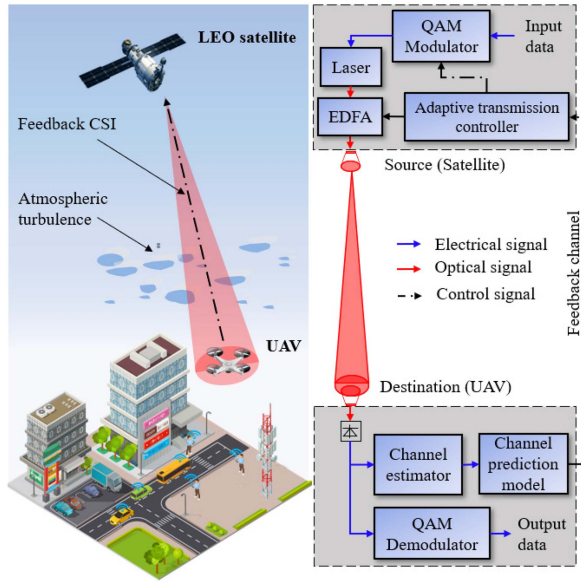


Fig. 1. FSO-based satellite-to-UAV system block diagram.

- C_1 : It is a design of an efficient adaptive modulation and power (AMP) control scheme for FSO-based LEO satellite systems. Particularly, we introduce the sub-optimal adaptive modulation and power (SAMP) scheme, enabling the LEO satellite to adjust the rate and power in the fixed equal-time slots. The design proposal guarantees a targeted achievable rate requested by the user while minimizing power consumption.
- C_2 : Unlike the single-output ESN channel prediction model reported in [24] and [26], we present a multistep-prediction ESN, where the predicted CSI effectively overcomes the induced feedback delay. Moreover, a comprehensive comparison with state-of-the-art ML-based channel prediction models in terms of accuracy, computational time, and energy efficiency performance is also included.
- C_3 : We provide insightful numerical results into the detailed impacts of turbulence channel conditions and the outdated CSI on the performance of our proposed rate/power adaptation system. In addition, we conduct the simulations to verify the correctness of the model and analysis.

The rest of this article is organized as follows. The system and channel models are described in Section II. Section III presents our proposed adaptive rate with a power control scheme and derives performance metrics, including average transmitted power and energy efficiency. Numerous numerical results and discussions are given in Section IV. Finally, Section V concludes this article.

II. SYSTEM AND CHANNEL MODELS

A. System Description

Fig. 1 illustrates an FSO transmission from an LEO satellite to an unmanned aerial vehicle (UAV), e.g., for vertical backhaul solutions or IoV applications. A joint

power and rate adaptation scheme is employed at the PHY layer. This aims to satisfy a specific data rate demand from the UAV with minimum power consumption under various practical constraints. For the adaptive rate transmission, we adopt the subcarrier K -ary quadrature amplitude modulation (K -QAM) scheme with a fixed symbol rate of R_s for M possible transmission modes. The transmission bit rate varies for each transmission mode and is given as $R_b = R_s \log_2(K)$ (bits/s), where K is the constellation size. Let $\mathbf{h}^* = \{h_1^* < h_2^* < \dots < h_M^* < h_{M+1}^* = \infty\}$ be the switching thresholds for M different transmission modes, $\mathbf{h} = \{h_1 < h_2 < \dots < h_N < h_{N+1} = \infty\}$ be the switching thresholds for N channel states, and h is the instantaneous channel gain, i.e., the CSI. The i th transmission mode is selected if $h_i^* \leq h < h_{i+1}^*$, and the current channel is said to be in the j th state if $h_j \leq h < h_{j+1}$, where $i \in \{1, 2, \dots, M\}$ and $j \in \{1, 2, \dots, N\}$. To avoid a high BER, the system falls into outage mode (no transmission) when the channel is in bad condition, i.e., $h < h_1 = h_1^*$. The selection of the transmission mode thresholds $\mathbf{h}^*$ and channel state thresholds $\mathbf{h}$ can be found in Section III.

On the other hand, the data bursts are transmitted in the equal duration of channel states. Each channel state is designed to cover a data burst transmission, in which its duration, denoted as T_b , is chosen to be shorter than the channel coherence time. This guarantees that the channel remains stable during a transmission. Here, both data rate and transmitted power are changed for each channel state using different transmission modes. At the UAV-based receiver side,² we employ an ESN-based prediction model whose purpose is to forecast the future CSI from several historical values. It is then fed back to the LEO satellite's transmitter. Based on this predicted CSI, the adaptive transmission controller decides the proper channel state and transmission mode of the system. It sends a control signal to the QAM modulator and the erbium-doped fiber amplifier (EDFA)³ to select the proper constellation size and amplifier gain for the subsequent burst transmission.

B. FSO Channel Model

For the FSO link between the LEO satellite and the UAV, we consider three major impairments,⁴ i.e., atmospheric attenuation h_1 , atmospheric turbulence h_a , and pointing

²The CSI is estimated at the UAV, which is then fed back to the satellite via feedback links, which could be an FSO or RF link. In FSO-based LEO satellite systems, the feedback delay (a few milliseconds) is much longer than the channel coherence time (typically shorter than 1 ms), leading to the outdated CSI [22]. Here, the imperfect CSI due to other sources, e.g., channel estimation and quantization errors, is beyond the scope of this study and can be further investigated in our future work.

³The typical time scale needed for changing EDFA gains is in the order of tens up to hundreds of microseconds [27], [28]. These values are, nevertheless, much smaller than the channel coherence time and the considered time slot (orders of milliseconds). Therefore, the EDFA delay does not have a significant impact on our considered system and thus can be ignored.

⁴As reported in [29, Sec. 2.1], the Doppler effect can be ignored as its maximum value is within the capability of the current receiver design for FSO-based satellite systems.

error h_p . The composite channel coefficient is expressed as $h = h_a h_l h_p$.

1) *Atmospheric Attenuation and Turbulence*: The molecular absorption and aerosol scattering suspended in the air significantly decrease received signal power. This is described by the Beer–Lambert law as $h_l = \exp(-\sigma(H_a - H_U)\sec(\xi))$, where σ is the attenuation coefficient related to the visibility V , while H_a , H_U , and ξ are the atmospheric altitude, UAV's altitude, and zenith angle, respectively [29, eq. (3)].

On the other hand, the atmospheric turbulence phenomenon causes the scintillation effect, resulting in signal power fluctuations at the UAV's detector. The UAVs considered in this article are drones, which operate within 1 km of the earth's surface. As a result, the turbulence strength for the LEO satellite-to-UAV optical links is usually weak [29]. The intensity fluctuations can be well described by a log-normal model, i.e., [8, eq. (6)]

$$f_{h_a}(h_a) = \frac{1}{h_a \sigma_R \sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{\ln h_a + \sigma_R^2/2}{\sigma_R}\right)^2\right] \quad (1)$$

where σ_R^2 is the Rytov variance determined as [4, eq. (15)]

$$\sigma_R^2 = 2.25 k_{\text{wave}}^{7/6} \sec^{11/6}(\xi) \int_{H_a}^{H_u} C_n^2(h) (h - H_u)^{5/6} dh \quad (2)$$

where $k_{\text{wave}} = 2\pi/\lambda$ is the optical wave number corresponding to the optical wavelength λ , and H_a is the atmospheric altitude. Also, $C_n^2(h)$ is given by the Hufnagel–Valley model as [3] $C_n^2(h) = 0.00594 \left(\frac{v_{\text{wind}}}{27}\right)^2 (10^{-5}h)^{10} \exp(-\frac{h}{1000}) + 2.7 \times 10^{-16} \exp(-\frac{h}{1500}) + C_n^2(0) \exp(-\frac{h}{100})$, where $C_n^2(0)$ is the ground-level turbulence and v_{wind} is the rms wind speed.

In addition, the aperture averaging effect demonstrates that an increase in the receiver aperture size leads to a decrease in power fluctuations caused by atmospheric turbulence. The aperture averaging factor, denoted as AA, is given as [29, eq. (9)]

$$AA = \frac{\sigma_1^2(d_a)}{\sigma_1^2(d_a = 0)} = \left[1 + 1.11 \left(\frac{d_a^2}{\lambda \Phi \sec(\xi)}\right)^{7/6}\right]^{-1} \quad (3)$$

where d_a is the aperture diameter, σ_1^2 is the scintillation index [3], and Φ is found in [29, eq. (10)].

2) *UAV Hovering-Induced Misalignment*: The misalignment between the center of the satellite's beam footprint and that of the UAV's detector is due to: 1) the satellite's vibration and the UAV's hovering at their own fixed positions and 2) the UAV's operation within the satellite's wide beam coverage.

The generalized misalignment model is reported in [29, Sec. 2.4]. The total radial displacement due to UAV hovering and satellite vibrations is given as $\mathbf{r} = (r_x, r_y)$, where $r_x \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $r_y \sim \mathcal{N}(\mu_y, \sigma_y^2)$. Here, r_x and r_y are two independent Gaussian random variables with different nonzero means (μ_x, μ_y) and variances (σ_x^2, σ_y^2) . Therefore, $\mathbf{r}$ follows Beckmann distribution, which is accurately approximated by a modified Rayleigh distribution [4,

eq. (17)], i.e.,

$$f_r(r) \approx \frac{r}{\sigma_m^2} \exp\left(-\frac{r^2}{2\sigma_m^2}\right), \quad r > 0 \quad (4)$$

where $\sigma_m^2 = \left(\frac{3\mu_x^2\sigma_x^4 + 3\mu_y^2\sigma_y^4 + \sigma_x^6 + \sigma_y^6}{2}\right)^{1/3}$ is the approximated jitter variance.

Considering the Gaussian beam profile, the fraction of received power at the UAV's detector is given as [19, eq. (9)]

$$h_p \approx A_0 \exp\left(-\frac{2r^2}{\omega_{\text{zeq}}^2}\right) \quad (5)$$

where $A_0 = [\text{erf}(v)]^2$ is the fraction of collected power at $r = 0$, $\text{erf}(\cdot)$ is the error function, $v = \frac{\sqrt{\pi}d_a}{2\sqrt{2}\omega_z}$ is the ratio between the aperture diameter and the beamwidth. In addition, $\omega_{\text{zeq}}^2 = \omega_z^2 \frac{\sqrt{\pi}\text{erf}(v)}{2v \exp(-v^2)}$ is the equivalent beamwidth, where ω_z is the effective beamwidth at distance L , given as $\omega_z = \omega_0 \sqrt{(\Delta_0^2 + \Lambda_0^2)(1 + 1.625\sigma_R^{12/5})}$, where $\omega_0 = \frac{2\lambda}{\pi\theta}$ is the beamwidth at $L = 0$, with θ being the divergence angle. Also, $\Delta_0 = 1 - \frac{L}{F_0}$, where F_0 is the radius of curvature, while $\Lambda_0 = \frac{2L}{k_{\text{wave}}\omega_0^2}$ and $\Lambda_1 = \frac{\Lambda_0}{\Delta_0^2 + \Lambda_0^2}$.

From (4) and (5), the probability distribution function (PDF) of h_p is derived as [29, eq. (15)]

$$f_{h_p}(h_p) = \frac{\varphi_m^2}{h_p} \left(\frac{h_p}{A_m}\right)^{\varphi_m^2}, \quad 0 \leq h_p \leq A_m \quad (6)$$

where $A_m = A_0 \exp\left(\frac{1}{\varphi_m^2} - \frac{1}{2\varphi_x^2} - \frac{1}{2\varphi_y^2} - \frac{\mu_x^2}{2\sigma_x^2\varphi_x^2} - \frac{\mu_y^2}{2\sigma_y^2\varphi_y^2}\right)$ and $\varphi_m = \frac{\omega_{\text{zeq}}}{2\sigma_m}$, with $\varphi_x = \frac{\omega_{\text{zeq}}}{2\sigma_x}$ and $\varphi_y = \frac{\omega_{\text{zeq}}}{2\sigma_y}$ the jitter variances in the x and y directions, respectively.

3) *Composite Channel Statistical Model*: The composite PDF of the channel gain $h = h_a h_l h_p$ is derived as [8, eq. (12)]

$$f_h(h) = \frac{\varphi_m^2 h^{\varphi_m^2-1}}{2(A_m h_l)^{\varphi_m^2}} \text{erfc}\left(\frac{Z}{\sqrt{2}\sigma_R}\right) \exp(0.5\sigma_R^2\varphi_m^2(1 + \varphi_m^2)) \quad (7)$$

where $Z = \log(\frac{h}{A_m h_l}) + \mu$ with $\mu = 0.5\sigma_R^2(1 + 2\varphi_m^2)$. The cumulative distribution function of h is given as [29, eq. (34)]

$$F_h(h) = \frac{1}{2} \exp[\varphi_m^2(Z - 0.5\sigma_R^2\varphi_m^2)] \text{erfc}\left(\frac{Z}{\sqrt{2}\sigma_R}\right) + \frac{1}{2} \text{erfc}\left(\frac{\sigma_R\varphi_m^2}{\sqrt{2}} - \frac{Z}{\sqrt{2}\sigma_R}\right). \quad (8)$$

C. ESN-Based Channel Prediction

1) *Multistep-Prediction ESN*: The multiple-input-multiple-output ESN structure is illustrated in Fig. 2. Particularly, the model consists of three different layers, i.e., an input layer $[1; \mathbf{x}(n)] \in \mathbb{R}^{(\mathcal{M}+1) \times 1}$, a hidden layer (reservoir) $\mathbf{r}(n) \in \mathbb{R}^{\mathcal{L} \times 1}$, and an output layer $\mathbf{y}(n) \in \mathbb{R}^{\mathcal{N} \times 1}$. The input layer is connected to the hidden reservoir by an input weight matrix $\mathbf{W}_i \in \mathbb{R}^{\mathcal{L} \times (\mathcal{M}+1)}$, while the neurons in the reservoir are mutually connected by an internal sparse matrix $\mathbf{W} \in \mathbb{R}^{\mathcal{L} \times \mathcal{L}}$. The updated and output equations are typically

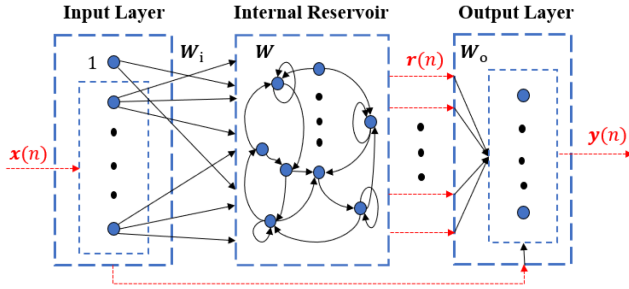


Fig. 2. Multiple-input-multiple-output ESN model.

defined as

$$\tilde{\mathbf{r}}(n) = \tanh(W_i[1; \mathbf{x}(n)] + W\mathbf{r}(n-1)) \quad (9)$$

$$\mathbf{r}(n) = (1 - \alpha)\mathbf{r}(n-1) + \alpha\tilde{\mathbf{r}}(n) \quad (10)$$

$$\mathbf{y}(n) = W_o[1; \mathbf{x}(n); \mathbf{r}(n)] \quad (11)$$

where $\tilde{\mathbf{r}}(n)$ is the updated state of $\mathbf{r}(n)$, $[\cdot; \cdot]$ is the vertical matrix concatenation, $\alpha \in (0, 1]$ is the leaking rate, and $W_o \in \mathbb{R}^{N \times (1+M+L)}$ is the output weight matrix.

2) *Channel Prediction Process*: We first prepare the dataset for training and testing the model. At discrete time n , M data samples are regarded as the input data, while the output is defined as the next N channel gain samples as

$$\begin{cases} \mathbf{x}(n) = [h(n\tau_s); h((n+1)\tau_s); \dots; h((n+M-1)\tau_s)] \\ \mathbf{y}(n) = [h((n+M)\tau_s); \dots; h((n+M+N-1)\tau_s)] \end{cases} \quad (12)$$

where τ_s denotes the sampling interval. Then, we start training the model to obtain the output weight matrix, as follows.

Step 1: Randomly generate W_i and W between 0 and 1. Then, take M consecutive channel gain as the input vector $\mathbf{x}(n)$; the state of the internal neurons $\mathbf{r}(n)$ is updated based on (10). The first N_i samples are used to initialize the reservoir state. The collected states are stored in a matrix $X \in \mathbb{R}^{(1+M+L) \times (N_i - N_i + 1)}$, given as

$$X = [Z(N_i), Z(N_{i+1}), \dots, Z(N_t)] \quad (13)$$

where $Z(n) = [1; \mathbf{x}(n); \mathbf{r}(n)]$, and N_t is the number of training samples. The corresponding target output matrix is $Y \in \mathbb{R}^{N \times (N_t - N_i + 1)}$, given by

$$Y = \begin{bmatrix} y(N_i) & y(N_i+1) & \dots & y(N_t) \\ y(N_i+1) & y(N_i+2) & \dots & y(N_t+1) \\ \vdots & \vdots & \ddots & \vdots \\ y(N_i+N-1) & y(N_i+N) & \dots & y(N_t+N-1) \end{bmatrix}. \quad (14)$$

Step 2: Based on X and Y , the output weight matrix can be obtained by using ridge regression as

$$W_o = YX^T (XX^T + \beta I)^{-1} \quad (15)$$

where β is the regularization coefficient and I is the identity matrix.

Finally, after obtaining the output weight matrix W_o , at discrete time n , N future CSI $\mathbf{y}(n)$ can be predicted based on M input samples $\mathbf{x}(n)$ according to (9)–(12).

III. ADAPTIVE TRANSMISSION SCHEMES

In our adaptive system, the modulation size K and transmitted power P_t adaptively vary according to the channel conditions. This aims to satisfy an achievable rate $\bar{r}$ requested from the UAV with minimum power consumption and meet practical constraints, i.e., targeted outage probability ($\text{Pr}_{\text{out,tar}}$), targeted BER (BER_{tar}), and maximum transmitted power ($P_{t,\max}$). The optimization problem is formulated as

$$\min_{P_t(h), K(h)} \bar{P}_t = \int_0^\infty P_t(h) f_h(h) dh \quad (16a)$$

$$\text{s.t. } \bar{r} = \int_0^\infty R_s \log_2(K(h)) f_h(h) dh \quad (16b)$$

$$\text{Pr}_{\text{out}} \leq \text{Pr}_{\text{out,tar}} \quad (16c)$$

$$\text{BER}(h) \leq \text{BER}_{\text{tar}} \quad (16d)$$

$$P_t(h) \leq P_{t,\max}. \quad (16e)$$

We focus on two adaptive schemes: 1) *ideal optimal AMP* [16], and 2) *our proposed SAMP*. Hereinafter, the modulation size and transmit power are denoted as functions of the channel coefficient h , i.e., $K(h)$ and $P_t(h)$, respectively.

A. AMP Scheme

In this ideal approach, the QAM constellation size K is discretely selected from $\mathbf{K} = \{K_1, K_2, \dots, K_M\}$, while the transmitted power P_t is continuously varied according to the channel conditions. Since K takes discrete values, the optimization stated in (16) is an integer programming [16]. To simplify it, we divide the channel gains h into subintervals, so-called transmission thresholds h^* . Each interval corresponds to one modulation size K_i . Let $\mathbf{P}_r = \{P_{r,1}, P_{r,2}, \dots, P_{r,M}\}$ be the received power thresholds based on $\mathbf{K}$ that guarantees BER_{tar} ; the optimal transmitted power can be expressed as

$$P_t(h) = \begin{cases} \frac{P_{r,i}}{h}, & h_i^* \leq h < h_{i+1}^*, i \in \{1, 2, \dots, M\} \\ 0, & h < h_1^* \end{cases} \quad (17)$$

where h_i^* is the outage threshold.

From (16c), h_1^* can be obtained by keeping the outage probability equal to $\text{Pr}_{\text{out,tar}}$, i.e.,

$$\text{Pr}_{\text{out,tar}} = \int_0^{h_1^*} f_h(h) dh = F_h(h_1^*). \quad (18)$$

The optimization problem is now reduced to determine the optimal transmission thresholds h^* and is written as

$$\min_{h^*} \bar{P}_t = \sum_{i=1}^M P_{r,i} \int_{h_i^*}^{h_{i+1}^*} \frac{f_h(h)}{h} dh \quad (19a)$$

$$\text{s.t. } \bar{r} = \sum_{i=1}^M R_s \log_2(K_i) \int_{h_i^*}^{h_{i+1}^*} f_h(h) dh \quad (19b)$$

$$h_1^* \leq h_2^* \leq \dots \leq h_M^*. \quad (19c)$$

Algorithm 1: AMP Scheme.

Input: $\bar{\tau}$, $\text{Pr}_{\text{out,tar}}$, $\mathbf{P}_r$, $\mathbf{K}$, h **Output:** $P_t(h)$, $K(h)$, $\mathbf{h}^*$ **Step 1:** Given $\text{Pr}_{\text{out,tar}}$, compute h_1^* from (18)**Step 2:** Given $\bar{\tau}$, compute $\mathbf{h}^*$ from (20)**Step 3:****if** $h \geq h_1^*$ **and** $h_i^* \leq h < h_{i+1}^*$ **then** **3.1:** Allocate optimal transmit power $P_t(h) = \frac{P_{r,i}}{h}$ **3.2:** Choose modulation order K_i **else**

Outage mode occurs, and the system is halted

end if

LEMMA 1 An optimum set of channel coefficient $\mathbf{h}^*$ can be derived as

$$h_i^*(\zeta) = \frac{1}{\zeta} \frac{P_{r,i} - P_{r,i-1}}{\log_2 K_i - \log_2 K_{i-1}}, \quad i \in \{2, \dots, M\} \quad (20)$$

where ζ is the Lagrange multiplier, which is obtained from

$$\begin{aligned} \bar{\tau} &= \sum_{i=1}^M R_s \log_2(K_i) \int_{h_i^*(\zeta)}^{h_{i+1}^*(\zeta)} f_h(h) dh \\ &= \sum_{i=1}^M R_s \log_2(K_i) (F_h(h_{i+1}^*(\zeta)) - F_h(h_i^*(\zeta))). \end{aligned} \quad (21)$$

PROOF It is solved by using the Lagrangian method provided in Appendix A. ■

The summary of the AMP is given in Algorithm 1.

REMARK 1 The AMP scheme theoretically requires the transmitted power to continuously adapt its value in a single-bit duration. This, however, poses critical challenges in FSO-based satellite communications due to high feedback delays and low channel coherence time.

B. Proposed SAMP Scheme

In this scheme, the transmitted power is selected from a list of discrete values $\mathbf{P}_t = [P_{t,1}, P_{t,2}, \dots, P_{t,N}]$. As mentioned in Section II, data are transmitted in fixed-time bursts. For each burst transmission, we need to determine the appropriate transmitted power $P_{t,j}$, $j \in \{1, 2, \dots, N\}$ and constellation size K_i , $i \in \{1, 2, \dots, M\}$. To do so, we partition the thresholds into two types of subintervals, i.e., mode thresholds $\mathbf{h}^*$ and channel state thresholds $\mathbf{h}$. Each modulation scheme K_i represents a transmission mode, while each transmitted power $P_{t,j}$ corresponds to a channel state.

REMARK 2 To implement the SAMP scheme, we need to: 1) find $\mathbf{h}^*$ and $\mathbf{h}$ and 2) determine the relation between these thresholds. Here, $\mathbf{h}^*$ can be obtained by solving the optimization problem stated in (19). Meanwhile, to find $\mathbf{h}$, we need to design a channel state model to effectively facilitate the system operation over the time-varying turbulence channels.

Algorithm 2: Channel State Threshold Determination.

Input: h_1^* **Output:** $\mathbf{h}$, N **Step 1:** Set $h_1 = h_1^*$, and $j = 2$ **Step 2:****while** true **do** **2.1:** Search h_j that satisfies $\bar{t}_j = T_b$, where $\bar{t}_j$ is given in (22) **2.2:** **if** $h_j - h_{j-1} \leq \epsilon_{\text{th}}$ **then** $h_j = \text{INF}$

Break

end if **2.2:** Set $j = j + 1$ **end while****Step 3:** Set $N = j - 1$

1) *Channel State Model:* In our system, the data are transmitted in bursts within the fixed-time slots of T_b . As a result, we divide the channel into nonoverlapping states defined by a range of channel coefficients. The selection of this range satisfies the condition that the intervals of all channel states are equal to T_b and shorter than the channel coherence time. The interval of the i th state, $\bar{t}_i$, depends on the statistical characteristic of the channel, which is given as [8, eq. (15)]

$$\bar{t}_j = \frac{\text{Pr}_j}{\text{LCR}(h_j) + \text{LCR}(h_{j+1})} \quad (22)$$

where $\text{Pr}_j = F_h(h_{j+1}) - F_h(h_j)$ is the probability of the i th channel state. Also, $\text{LCR}(h_{\text{th}})$ is the level crossing rate at a certain threshold of h_{th} given as [30, (37)]

$$\begin{aligned} \text{LCR}(h_{\text{th}}) &= \frac{\omega_{\text{zeq}}}{8\pi\sigma_m^2\sigma_R} \sum_{k=1}^J w_k \sqrt{8\eta^2 x_k + b\sigma_R^2\omega_{\text{zeq}}} \\ &\times \exp\left(-\frac{(x_k + \ln(h_{\text{th}}/\Upsilon) + \sigma_R^2/2)^2}{2\sigma_R^2} + x_k(1 - \varphi_m^2)\right) \end{aligned} \quad (23)$$

where $\eta^2 = 2\sigma_m^2|\Omega^2 - \frac{\Gamma^2}{4}|$, with Ω and Γ being found in [30], and $b = 2\pi\sigma_c^2$ with $\sigma_c = \frac{f_c}{\sqrt{2\ln 2}}$; f_c is the 3-dB cutoff frequency. Also, w_k and x_k are, respectively, the weight factor and the k th zero of the Laguerre polynomials found in [31, Tab. 25.9], and J is the number of Gauss–Laguerre approximation orders.

2) *SAMP Algorithm:* To determine channel state thresholds $\mathbf{h}$, we use the same approach stated in [8]. Specifically, we set the first threshold h_1 to be equal to h_1^* to keep $\text{Pr}_{\text{out,tar}}$, and the channel state interval $\bar{t}_j$ to be equal to the burst duration T_b . In this way, all threshold levels $\{h_i\}_{i=1}^{N+1}$ are determined. The detailed approach is summarized in Algorithm 2, where ϵ_{th} is the gap threshold between two consecutive channel states.

Algorithm 3: Relation Vector Determination.

Input: h^*, h
Output: A
Step 1: Set $j = 1$
Step 2:
while $j \leq N$ **do**
 2.1: Find $i, i \in \{1, 2, \dots, M\}$, such that $h_j \geq h_i^*$
 and $h_j < h_{i+1}^*$
 2.2: Set $A_j = i$
 2.2: Set $j = j + 1$
end while

Algorithm 4: SAMP Scheme.

Input: $\bar{\tau}, \text{Pr}_{\text{out,tar}}, P_r, K, h$
Output: $P_t(h), K(h), h^*, h, A, P_t$
Step 1: Given $\text{Pr}_{\text{out,tar}}$, compute h_1^* from (18)
Step 2: Given $\bar{\tau}$, compute h^* from (20)
Step 3: Compute h according to Algorithm 2
Step 4: Compute A by assigning the transmission modes to channel states according to Algorithm 3
Step 5: Compute P_t from (24)
Step 6:
if $h \geq h_1$ **and** $h_j \leq h < h_{j+1}$ **then**
 6.1: Choose transmit power $P_{t,j}$
 6.2: Choose modulation order K_{A_j}
else
 Outage mode occurs, and the system is halted
end if

When all channel states and transmission modes are determined, the next step is to find the relation between these thresholds. Let $A = [A_1, A_2, \dots, A_N]$, where A_j is the transmission mode of the j th channel state. In general, one channel state interval is usually smaller than that of a transmission mode. If the mode i covers all the channel state j , we assign $A_j = i$. Otherwise, if the channel state j belongs to two or more transmission modes, the lowest mode is selected to guarantee the condition (16d), i.e., $\text{BER}(h) \leq \text{BER}_{\text{tar}}$.

Based on the relation vector A found in Algorithm 3, the transmitted power $P_{t,j}$ used for the j th channel state, $j \in \{1, 2, \dots, N\}$, can be expressed as

$$P_{t,j} = \frac{P_{r,A_j}}{h_j}. \quad (24)$$

The summary of the SAMP is given in Algorithm 4.

C. Performance Analysis

1) *Average Required Transmitted Power:* It is defined as the minimum transmitted power to maintain a particular requested rate $\bar{\tau}$ under predetermined constraints on $\text{Pr}_{\text{out,tar}}$, BER_{tar} , and $P_{t,\text{max}}$.

LEMMA 2 The closed-form expression of required transmit power for the AMP scheme can be expressed as

$$\begin{aligned} \bar{P}_{t,\text{AMP}} = & \sum_{i=1}^M \frac{\varphi_m^2 \vartheta_m P_{r,i}}{2A_m h_i B'} \left[\exp(y_{i+1} B') \text{erfc}(y_{i+1} A') \right. \\ & - \exp(y_i B') \text{erfc}(y_i A') - \exp(B'^2 / 4 A'^2) \\ & \left. \times \left(\text{erf}\left(\frac{B'}{2A'} - y_{i+1} A'\right) - \text{erf}\left(\frac{B'}{2A'} - y_i A'\right) \right) \right] \end{aligned} \quad (25)$$

where $\vartheta_m = \exp(\sigma_R^2(0.5 + \varphi_m^2 - 0.5\varphi_m^4))$, $A' = \frac{1}{\sqrt{2}\sigma_R}$, $B' = \varphi_m^2 - 1$, $y_i = \log(\frac{h_i^*}{A_m h_i}) + \mu$, and $y_{i+1} = \log(\frac{h_{i+1}^*}{A_m h_i}) + \mu$.

PROOF See Appendix B. ■

On the other hand, the average transmitted power of the SAMP scheme can be computed as

$$\begin{aligned} \bar{P}_{t,\text{SAMP}} &= \sum_{j=1}^N P_{t,j} \int_{h_j}^{h_{j+1}} f_h(h) dh \\ &= \sum_{j=1}^N P_{t,j} (F_h(h_{j+1}) - F_h(h_j)). \end{aligned} \quad (26)$$

2) *Energy Efficiency:* It is defined as the ratio between the average number of successfully received data bits within a burst duration and the average energy consumption for a burst transmission. It is, then, given as $\bar{\eta}_{\text{EE}} = \bar{E}_b / (\bar{P}_t \times T_b)$, where $\bar{E}_b$ is the average number of correctly received data bits within a burst duration. For the AMP scheme, it can be determined as

$$\begin{aligned} \bar{E}_{b,\text{AMP}} &= \sum_{i=1}^M R_s T_b \log_2 K_i (1 - \text{BER}_{\text{tar}}) \int_{h_i^*}^{h_{i+1}^*} f_h(h) dh \\ &= \sum_{i=1}^M R_s T_b \log_2 K_i (1 - \text{BER}_{\text{tar}}) (F_h(h_{i+1}^*) - F_h(h_i^*)). \end{aligned} \quad (27)$$

In the case of the SAMP scheme, $\bar{E}_b$ is computed as

$$\begin{aligned} \bar{E}_{b,\text{SAMP}} &= \sum_{j=1}^N R_s T_b \log_2 K_{A_j} (1 - \overline{\text{BER}}_j) \int_{h_j}^{h_{j+1}} f_h(h) dh \\ &= \sum_{j=1}^N R_s T_b \log_2 K_{A_j} (1 - \overline{\text{BER}}_j) (F_h(h_{j+1}) - F_h(h_j)) \end{aligned} \quad (28)$$

where $\overline{\text{BER}}_j = \int_{h_j}^{h_{j+1}} \text{BER}_j(h) f_h(h) dh$ is the average BER of the j th channel state. Here, $\text{BER}_j(h)$ is the instantaneous BER given as $\text{BER}_j(h) = 0.2 \exp(\frac{-3P_{t,j}^2 h^2}{2\sigma_n^2(K_{A_j}-1)})$, where σ_n is the standard deviation of the Gaussian noise.

IV. NUMERICAL RESULTS AND DISCUSSIONS

This section presents and discusses the performance of our proposed SAMP scheme in terms of energy efficiency. The effectiveness of the proposed design is also highlighted by comparing its performance with the conventional AM

TABLE II
System Parameters

Name	Symbol	Value
LEO Satellite Parameters		
LEO satellite altitude	H_S	500 km
Zenith angle	ξ	50°
Divergence half-angle	θ	$223 \mu\text{rad}$
Jitter angle	θ_{jt}	$11.15 \mu\text{rad}$
Symbol rate	R_s	250 Ms/s
Optical wavelength	λ	$1.55 \mu\text{m}$
UAV Parameters		
UAV's altitude	H_U	100 m
Aperture diameter	d_a	10 cm
Noise standard deviation	σ_n	10^{-7} A/Hz
Initial radial displacement	ρ	0 m
Other Parameters		
Burst duration	T_b	1 ms
Requested rate	$\bar{r}$	1 Gb/s
Target BER	BER_{tar}	10^{-5}
Target outage probability	$\text{Pr}_{\text{out,tar}}$	1 %
Atmospheric altitude	H_a	20 km
Ground turbulence level	$C_n^2(0)$	$10^{-14} \text{ m}^{-2/3}$
Wind speed	v_{wind}	21 m/s
Visibility (clear sky)	V	20 km

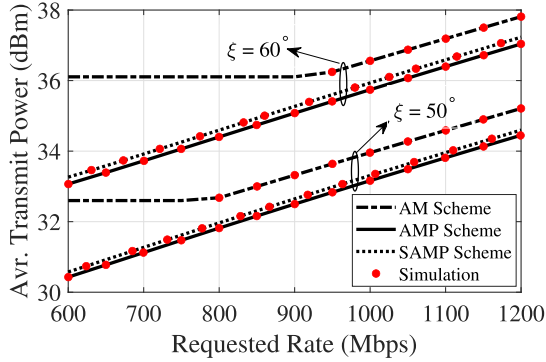


Fig. 3. Average required transmit power versus requested rate for different adaptive transmission schemes.

scheme [6] and ideal AMP scheme [16]. Moreover, the energy efficiency performance in the case of outdated CSI and predicted CSI using ML-based channel prediction models is also investigated. Monte Carlo simulations are performed to verify the analytical results. In addition, we adopt a set of constellation sizes $K = \{4, 8, 16, 32, 64, 128\}$. Other parameters used in this article, unless otherwise noted, are given in Table II. It is worth noting that most of parameters from Table II are adopted from [4], [30], and [32].

A. Average Transmitted Power and Energy Efficiency

First, we quantitatively highlight the effectiveness of the proposed SAMP scheme by investigating the required transmitted power to satisfy a particular rate, as depicted in Fig. 3. For the sake of comparison, in addition to the ideal AMP scheme [16], we consider the AM scheme, in which the transmitted power is fixed, and the constellation size is varied during the transmission [6]. Also, different satellite's zenith angles, i.e., $\xi = 50^\circ$ and 60° , are considered. As expected, the performance of the proposed SAMP is close to that of the ideal AMP scheme. Moreover, this scheme

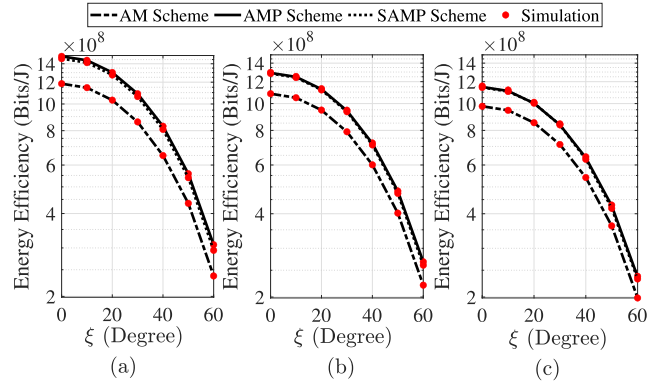


Fig. 4. Energy efficiency versus zenith angle for different requested rates. (a) $\bar{r} = 0.6$ Gb/s. (b) $\bar{r} = 1$ Gb/s. (c) $\bar{r} = 1.2$ Gb/s.

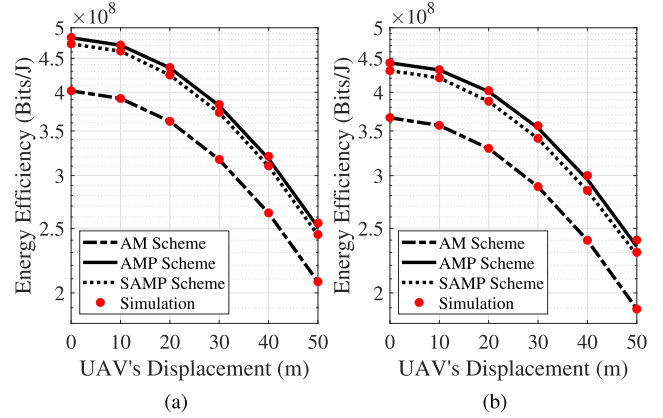


Fig. 5. Energy efficiency versus UAV's initial displacement for different adaptive transmission schemes. (a) $\theta_{jt} = 11.15 \mu\text{rad}$. (b) $\theta_{jt} = 20.07 \mu\text{rad}$.

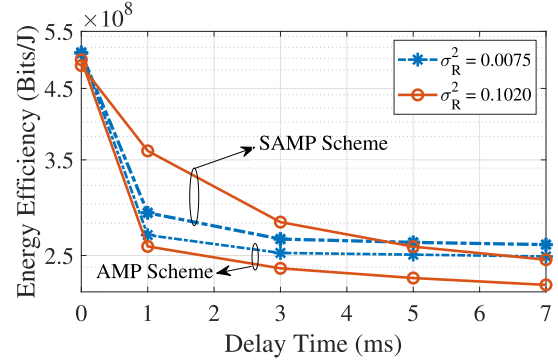


Fig. 6. Energy efficiency in the presence of outdated CSI.

also exhibits a significant performance enhancement at the cost of high complexity compared to the conventional AM scheme. For example, when $\xi = 50^\circ$ and requested rate = 1 Gb/s, the required power gap between AM and AMP/SAMP is approximately 0.85 dB. It is because the system can utilize joint AMP control to save power consumption. From this figure, the analytical results closely follow simulated ones, which validates the correctness of the model and analysis.

Fig. 4 analyzes the energy efficiency of different adaptive transmission schemes over a range of satellite's zenith

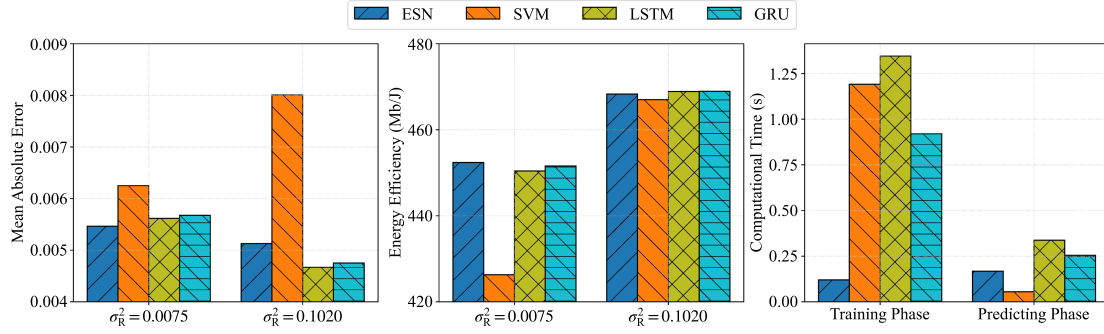


Fig. 7. Comparison of MAE, computational time, and energy efficiency performance for different channel prediction models.

angles. Also, different UAV's requested rates are considered, i.e., 0.6, 1, and 1.2 Gb/s. As seen, when the satellite's zenith angle increases, the energy efficiency performance experiences a considerable drop. This is because the satellite needs more power to maintain the QoS requirements, i.e., the requested rate and targeted outage probability, with higher atmospheric attenuation and stronger turbulence. In addition, when the requested rate by the UAV increases, energy efficiency decreases. In other words, there is a tradeoff between the throughput and energy efficiency of the system. For example, to offer a double requested rate, i.e., from 0.6 to 1.2 Gb/s, the system needs to sacrifice approximately 0.2 Gb/J in energy efficiency. This tradeoff comes from the fact that the power required to achieve a higher rate is on a larger scale than the increase in the rate itself.

Next, Fig. 5 analyzes the energy efficiency for different adaptive transmission schemes. Also, different pointing error conditions are considered, i.e., UAV's radial displacement as well as satellite vibration. As evident, an increase in the UAV's positions from the center of the beam footprint results in a significant decrease in energy efficiency, due to the beam spreading loss. For the AMP and SAMP schemes, energy efficiency is reduced by 200 Mb/J when the UAV is at 50 m from the beam center. Moreover, the severity of pointing error caused by satellite vibration jitter angle has a noticeable effect on energy efficiency due to frequent signal fluctuations. For instance, when the jitter angle varies from $\theta_{jt} = 11.15 \mu\text{rad}$ to $\theta_{jt} = 20.07 \mu\text{rad}$, energy efficiency decreases approximately 40 Mb/J.

B. Effects of Outdated and Predicted CSI

We now evaluate the system performance in the presence of outdated CSI with the channel prediction model. For the purpose of simulation and data source limitation, we adopt the channel data obtained from an FSO channel measurement [33]. We use two real-time FSO channel datasets corresponding to $\sigma_R^2 = 0.0075$ and $\sigma_R^2 = 0.1020$.

Fig. 6 investigates energy efficiency in the presence of delayed CSI for AMP and SAMP schemes. In this article, we assume that the delay time of the CSI is equal to nT_b , $n \in \mathbb{N}$, so that each burst transmission is only covered by a specific CSI. Obviously, the delayed CSI leads to a substantial degradation in energy efficiency. Moreover, as the

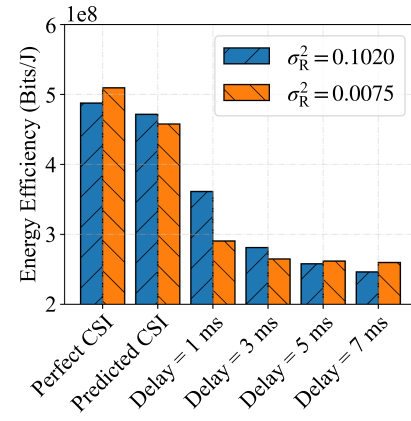


Fig. 8. Energy efficiency with predicted and outdated CSI.

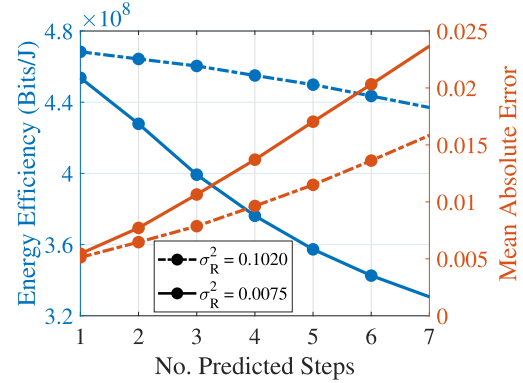


Fig. 9. Energy efficiency and MAE versus the number of predicted steps for the ESN model.

power is continuously varied in the optimal AMP scheme, it becomes more susceptible to imperfect CSI, leading to lower energy efficiency compared to the SAMP scheme. In addition, for the SAMP scheme, energy efficiency under stronger turbulence condition ($\sigma_R^2 = 0.1020$) is higher than that under weaker condition ($\sigma_R^2 = 0.0075$). This occurs because the interval between two consecutive channel state thresholds becomes narrower when the channel gets better. Consequently, it is more likely that the delayed state is different from the actual one, leading to high degradation in energy efficiency.

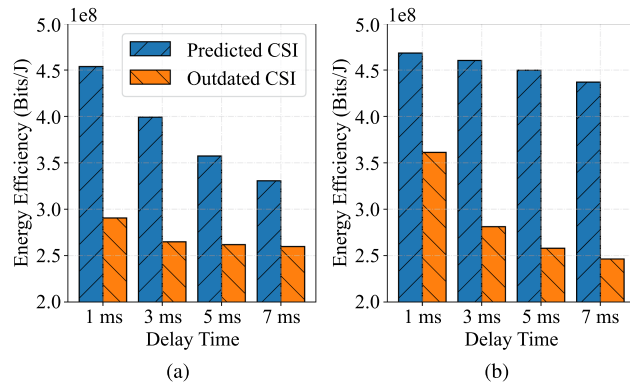


Fig. 10. Energy efficiency with outdated CSI and corresponding multistep predicted CSI. (a) $\sigma_R^2 = 0.0075$. (b) $\sigma_R^2 = 0.1020$

Next, we adopt a support vector machine (SVM) [34] and three different types of RNN, i.e., ESN [24], [26], long short-term memory (LSTM) [35], and gated recurrent unit (GRU) [36], for channel prediction. The total number of data samples is 837 500 with the sampling interval of 1 ms. For each iteration, we use 7500 samples for training and the next 2500 samples for testing the models. The numbers of input units, hidden neurons, and output units for RNN-based schemes are 20, 50, and 1, respectively. Meanwhile, the type of SVM is epsilon-SVR, the epsilon in the loss function is 0.01, and the kernel type is the radial basis function with the gamma value of 2.8. All simulations are performed in a Windows 11, AMD CPU Ryzen 7-5800H with 16.0 GB of RAM.

As shown in Fig. 7, the RNN-based prediction models outperform SVM in terms of mean absolute error (MAE) and energy efficiency, which confirms their effectiveness in processing the time-series data. In addition, the ESN model offers very close performance compared to LSTM and GRU while requiring much smaller training time thanks to its simple structure. This is an essential factor to consider since the FSO channel coherence time is in the order of milliseconds. Therefore, we select the ESN as the most potential prediction scheme to investigate the system's performance in this article.

In Fig. 8, we quantitatively compare the system's energy efficiency performance with perfect, predicted, and delayed CSI. As can be seen, in both cases of the channel, the system with predicted CSI offers very close energy efficiency compared to the system with perfect CSI. Furthermore, using predicted CSI provides a considerable performance enhancement in comparison with delayed CSI. In particular, energy efficiency when the ESN model is utilized is improved by approximately 110 Mb/J ($\sigma_R^2 = 0.1020$) and 160 Mb/J ($\sigma_R^2 = 0.0075$) compared to the 1-ms-delay CSI.

Next, we investigate the performance of the multistep-prediction ESN, in which the model predicts multiple future values from the previous ones, so that the predicted CSI totally overcomes the induced feedback delay. This is necessarily important for long-distance FSO-based satellite communications. In general, when increasing the number of predicted steps, there are an upward trend in the MAE

and an inverse trend for energy efficiency, as illustrated in Fig. 9. In addition, the ESN seems to perform better when $\sigma_R^2 = 0.1020$, which results in a lower MAE. It is worth noting that energy efficiency under weaker turbulence ($\sigma_R^2 = 0.0075$) suffers from a considerable decrease compared with stronger turbulence due to higher MAE and narrower channel state thresholds.

Finally, Fig. 10 compares the energy efficiency performance with outdated CSI and predicted CSI. We assume that using n -step prediction could completely overcome n ms of delay. As can be seen from the figure, energy efficiency with multistep prediction still experiences a significant improvement over that with outdated CSI, especially when $\sigma_R^2 = 0.1020$. This confirms that the channel prediction model is necessarily important for our AMP control system.

V. CONCLUSION

This article has introduced the design of the SAMP scheme with ML-based channel prediction. This scheme was confirmed to be more suitable for FSO-based LEO satellite communication systems than the conventional optimal ones. We analytically studied the system's average transmitted power and energy efficiency with different adaptive schemes and CSI conditions. There are several remarkable findings from the obtained results, as follows.

- 1) The proposed SAMP scheme exhibited a substantial performance enhancement at the cost of high complexity compared with the conventional AM scheme. Moreover, this scheme also offered very close performance compared with the ideal AMP. In the case of imperfect CSI, the ideal one was more susceptible to delayed feedback, leading to a considerable degradation of energy efficiency.
- 2) The ESN-based prediction model was utilized to cope with delayed CSI, thanks to its high accuracy and simple structure. A comprehensive comparison among different ML-based prediction models was also provided.
- 3) By using predicted CSI, the system experienced a considerable energy efficiency enhancement in comparison with delayed CSI. This highlighted the effectiveness of using ESN-based channel prediction for our proposed optical satellite system using adaptive transmission schemes.

Finally, Monte Carlo simulations further confirmed the correctness of the derived analytical results.

In addition, the study in this article envisions several future research directions, as follows.

- 1) Since this study focused on imperfect CSI due to the delayed feedback only, other impairment sources, such as channel estimation and quantization errors, would be an extended direction.
- 2) Another interesting future direction would be an investigation into the support of multiple UAVs, where each UAV suffers from different channel conditions.

In this scenario, designing an efficient resource allocation scheme that satisfies the QoS of each UAV while maintaining the fairness among them becomes a challenging problem.

APPENDIX A SOLVING THE OPTIMIZATION PROBLEM OF (19)

We use the Lagrange method to solve the optimization problem stated in (19), which can be formulated as

$$\begin{aligned} \mathcal{L}(\mathbf{h}^*, \zeta, \psi_1, \psi_2, \dots, \psi_M) \\ = - \sum_{i=1}^M P_{r,i} \int_{h_i^*}^{h_{i+1}^*} \frac{f_h(h)}{h} dh + \sum_{i=1}^M \psi_i (h_{i+1}^* - h_i^*) \\ + \zeta \left(\sum_{i=1}^M R_s \log_2(K_i) \int_{h_i^*}^{h_{i+1}^*} f_h(h) dh - \bar{\tau} \right) \end{aligned} \quad (29)$$

where $(\psi_1, \psi_2, \dots, \psi_M)$ and ζ are the Lagrange multipliers. Using the Karush–Kuhn–Tucker conditions [37], the optimal transmission thresholds $\mathbf{h}^*$, the corresponding Lagrange multipliers $(\psi_1, \psi_2, \dots, \psi_M)$, and ζ must satisfy the following conditions:

$$\frac{\partial}{\partial h_i^*} \mathcal{L}(\mathbf{h}^*, \zeta, \psi_1, \psi_2, \dots, \psi_M) = 0, \quad i \in \{2, \dots, M\} \quad (30a)$$

$$\psi_i \geq 0, \quad i \in \{2, \dots, M\} \quad (30b)$$

$$\psi_i (h_{i+1}^* - h_i^*) = 0. \quad (30c)$$

To simplify the optimization problem, we assume that $h_1^* < h_2^* < \dots < h_M^*$. Therefore, from (30b) and (30c), $\psi_i = 0$. By substituting (29) into (30a) and after some mathematical manipulations, the thresholds $\mathbf{h}^*$ can be derived as in (20).

We now confirm the accuracy of the abovementioned assumption. When all thresholds are determined, let us define the vector $\epsilon = [\epsilon_1, \dots, \epsilon_{M-1}]$, whose each element is computed as

$$\epsilon_i = \begin{cases} h_2^* - h_1^*, & i = 1 \\ \frac{P_{r,i+1} - P_{r,i}}{\log_2 K_{i+1} - \log_2 K_i} - \frac{P_{r,i} - P_{r,i-1}}{\log_2 K_i - \log_2 K_{i-1}}, & i \in \{2, \dots, M-1\}. \end{cases} \quad (31)$$

The assumption is valid if all the elements of ϵ are positive. Here, the values of ϵ_i only depend on the outage probability, received power threshold $P_{r,i}$, and modulation size K_i . As a result, a set of QAM constellation sizes and the outage probability are selected properly to guarantee the assumption. In the numerical results, we show that all the elements of ϵ are positive.

APPENDIX B DERIVATION OF (25)

The average transmitted power for the AMP is given as

$$\bar{P}_{t,AMP} = \sum_{i=1}^M P_{r,i} \int_{h_i^*}^{h_{i+1}^*} \frac{f_h(h)}{h} dh$$

$$\begin{aligned} = \sum_{i=1}^M P_{r,i} \frac{\varphi_m^2}{2(A_m h_l) \varphi_m^2} \exp(0.5 \sigma_R^2 \varphi_m^2 (1 + \varphi_m^2)) \\ \times \int_{h_i^*}^{h_{i+1}^*} h^{\varphi_m^2 - 2} \operatorname{erfc} \left(\frac{\log \left(\frac{h}{A_m h_l} \right) + \mu}{\sqrt{2} \sigma_R} \right) dh. \end{aligned} \quad (32)$$

Let $y = \log \left(\frac{h}{A_m h_l} \right) + \mu$; then, $h = A_m h_l \exp(y - \mu)$, $dh = A_m h_l \exp(y - \mu) dy$. Integral in (32) is rewritten as

$$\begin{aligned} \int_{h_i^*}^{h_{i+1}^*} h^{\varphi_m^2 - 2} \operatorname{erfc} \left(\frac{\log \left(\frac{h}{A_m h_l} \right) + \mu}{\sqrt{2} \sigma_R} \right) dh \\ = (A_m h_l)^{\varphi_m^2 - 1} \exp(-\mu (\varphi_m^2 - 1)) \\ \times \int_{y_i}^{y_{i+1}} \exp(y (\varphi_m^2 - 1)) \operatorname{erfc} \left(\frac{y}{\sqrt{2} \sigma_R} \right) dy \end{aligned} \quad (33)$$

where $y_i = \log \left(\frac{h_i^*}{A_m h_l} \right) + \mu$ and $y_{i+1} = \log \left(\frac{h_{i+1}^*}{A_m h_l} \right) + \mu$. Using [38, eq. (06.27.21.0011.01)], the integral in (33) is solved. After several mathematical manipulations, we complete the proof.

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